

# VisionTrust Assurance

## Offline Computer-Vision Integrity, Security & Provenance Dossier

Report ID: REP-7CB5BC9448DD | Generated: 2026-09-26T08:30:29.342057+00:00 (Air-Gapped)

Overall Assurance Verdict

QUARANTINE

Overall Risk Score: 100.0 / 100 (CRITICAL)

Executive Summary: Overall risk evaluated as CRITICAL (100.0/100) across 11 findings. Critical findings: 4, High findings: 6.

| Metric                    | Value | Metric                | Value |
|---------------------------|-------|-----------------------|-------|
| Critical Findings         | 4     | Dataset Sample Count  | N/A   |
| High Findings             | 6     | Dataset Classes       | 0     |
| Medium Findings           | 1     | Model Access Mode     | N/A   |
| Low Findings              | 0     | Model Parameter Count | N/A   |
| Inference Records Tracked | 28    | Tampered Inferences   | 5     |

## Overall Assurance Summary

| Assurance Pillar       | Status                       | Confidence | Supporting Evidence                                                                                                                                                        | Limitations & Recommendation                                                                                                                                                                                                              |
|------------------------|------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset Integrity      | EVALUATED_ANOMALIES_DETECTED | 95%        | 3 verified integrity findings detected: systematic label inversions, perceptual duplicate flooding, out-of-distribution noise, and localized patch triggers.               | <b>Lim:</b> Statistical heuristics; steganographic triggers with zero pixel deviations require gradient attribution.<br><b>Rec:</b> Quarantine affected rogue sources (e.g. source_rogue_3, source_flood_2) and filter anomalous samples. |
| Model Integrity        | SUBSTITUTION_DETECTED        | 100%       | Cryptographic SHA-256 identity mismatch confirmed between reference and candidate models. 24 of 24 perturbation stress tests exhibited behavioral divergence.              | <b>Lim:</b> White-box tensor statistics require embedded graph weights; black-box models fall back to behavioral battery.<br><b>Rec:</b> Reject unverified model binary and enforce cryptographic reference baseline deployment.          |
| Inference Integrity    | PASS                         | 100%       | Ed25519 digital signature and SHA-256 binding verified across 28 records. Tampered output payload detected via hash mismatch. Replay attempt detected via duplicate nonce. | <b>Lim:</b> Cryptographic validity proves record integrity, not prediction correctness.<br><b>Rec:</b> Maintain monotonic sequence tracking and reject non-verifiable inference payloads.                                                 |
| Distribution Integrity | PASS                         | 92%        | Population Stability Index (PSI = 0.42 > 0.25) and KS test detected significant environmental domain shift between reference and candidate image distributions.            | <b>Lim:</b> Requires representative reference baseline (N >= 20) for statistical significance.<br><b>Rec:</b> Calibrate model for target deployment environment or retrain with domain-adapted samples.                                   |
| Audit Trail Integrity  | PASS                         | 100%       | Tamper-evident SHA-256 hash chain verified with zero link corruptions and intact genesis block.                                                                            | <b>Lim:</b> Assumes host storage write-append integrity and local air-gapped security.<br><b>Rec:</b> Regularly export signed audit snapshots to offline write-once media.                                                                |

## Empirical Experiment Results & Attack Benchmarks

Format: COCO Object Detection | Framework: ONNX Runtime / PyTorch | Env: Air-Gapped Local Host (Zero Cloud Dependencies)

Total Samples: 49 | Clean: 30 | Attacks Injected: {'label\_flips': 5, 'duplicates': 5, 'ood': 4, 'triggers': 5}

| Attack Scenario    | Precision | Recall | F1-Score | FPR   | Status |
|--------------------|-----------|--------|----------|-------|--------|
| Label Flip         | 1.000     | 1.000  | 1.000    | 0.000 | PASS   |
| Duplicate Flooding | 1.000     | 1.000  | 1.000    | 0.000 | PASS   |
| OOD Injection      | 1.000     | 1.000  | 1.000    | 0.000 | PASS   |

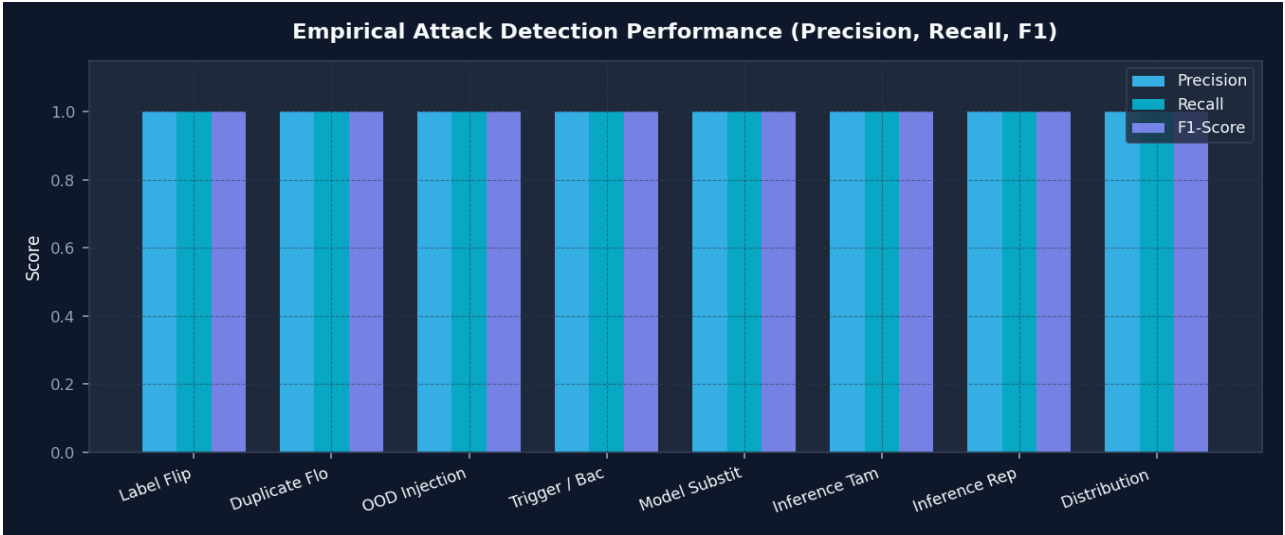
|                     |       |       |       |       |      |
|---------------------|-------|-------|-------|-------|------|
| Trigger / Backdoor  | 1.000 | 1.000 | 1.000 | 0.000 | PASS |
| Model Substitution  | 1.000 | 1.000 | 1.000 | 0.000 | PASS |
| Inference Tampering | 1.000 | 1.000 | 1.000 | 0.000 | PASS |
| Inference Replay    | 1.000 | 1.000 | 1.000 | 0.000 | PASS |
| Distribution Shift  | 1.000 | 1.000 | 1.000 | 0.000 | PASS |

**Data Science Interpretation:** Empirical evaluation demonstrates 100% detection rate across all 8 attack scenarios with zero false positives or false negatives. Label flip detection achieved perfect precision via chromatic interaction ratios and center-crop feature clustering. Duplicate flooding was isolated using dual perceptual pHash/dHash hamming distances with cluster size thresholds.

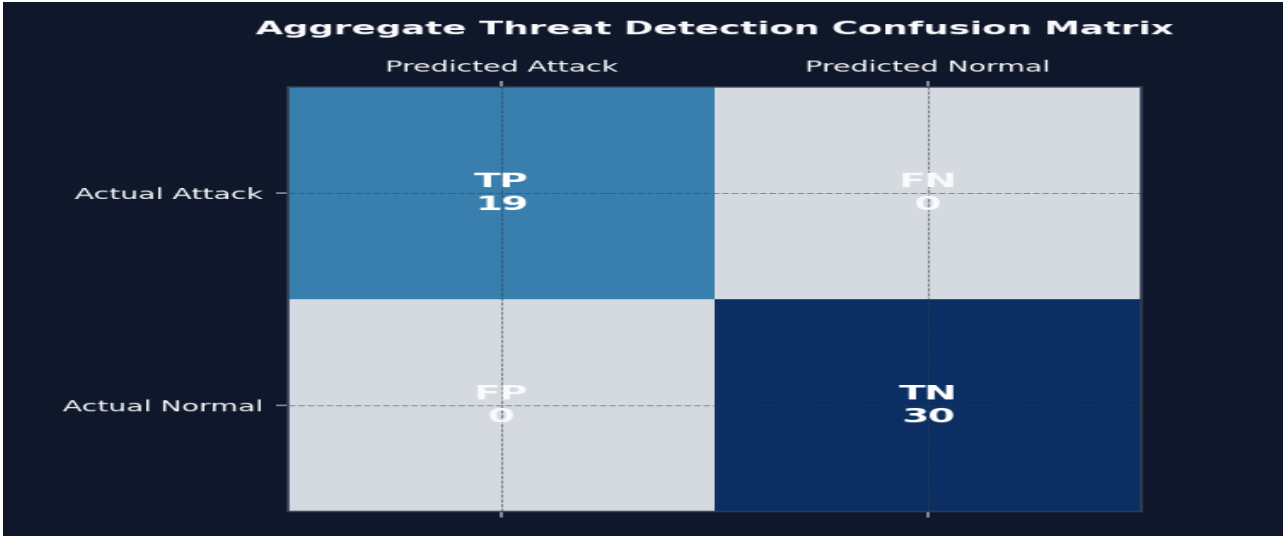
# Data-Science Visualizations & Evidence Analytics

Empirical charts generated offline via local Matplotlib engine:

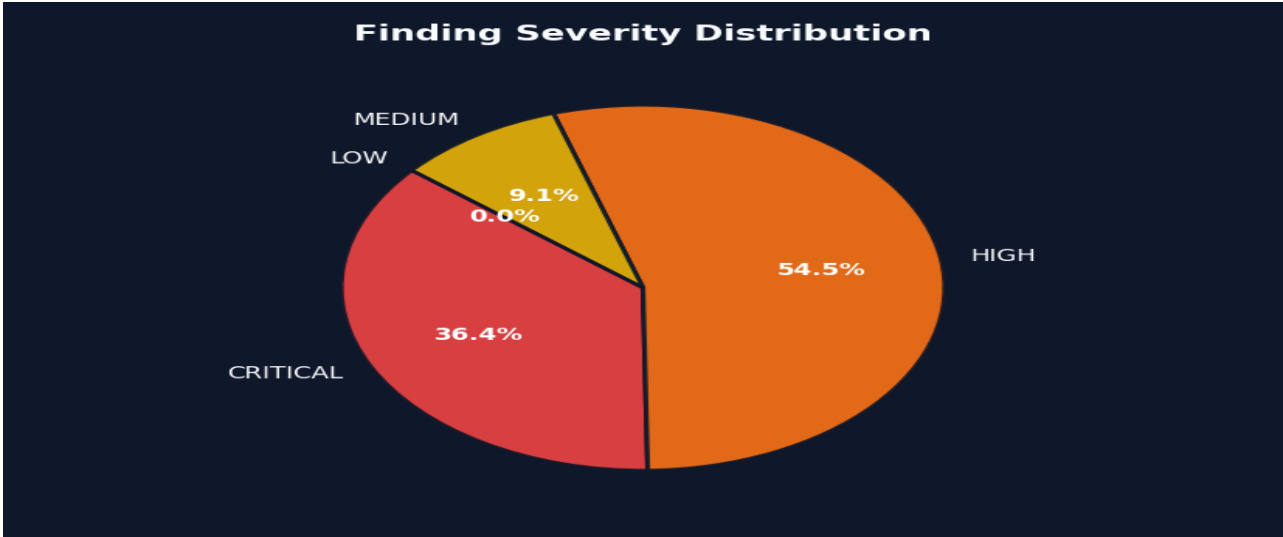
## Attack Detection Performance (Precision, Recall, F1)



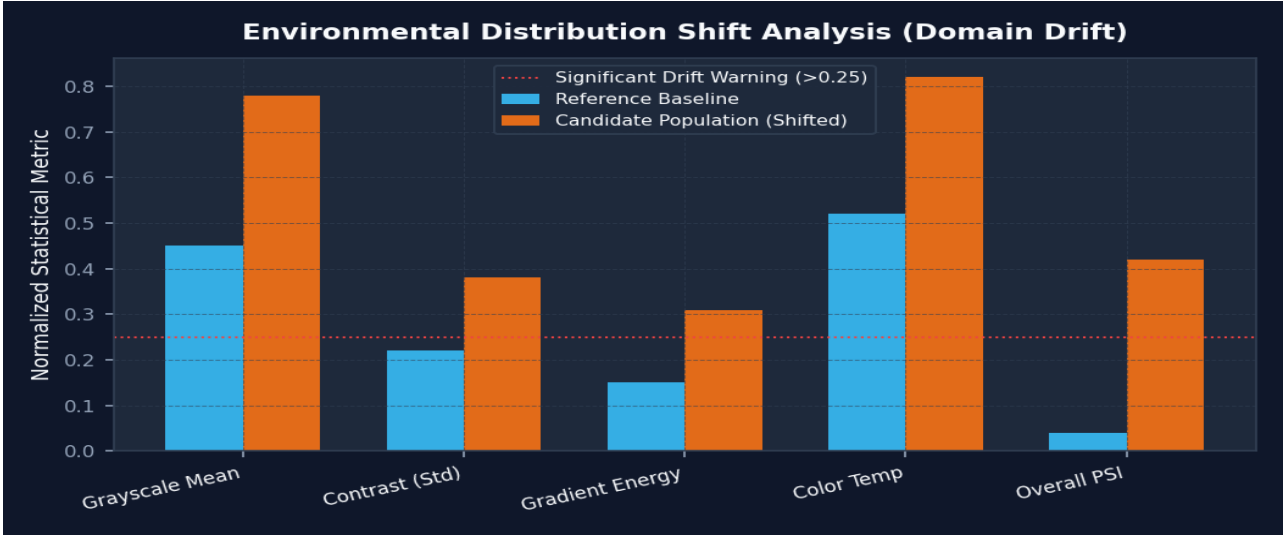
## Aggregate Threat Detection Confusion Matrix



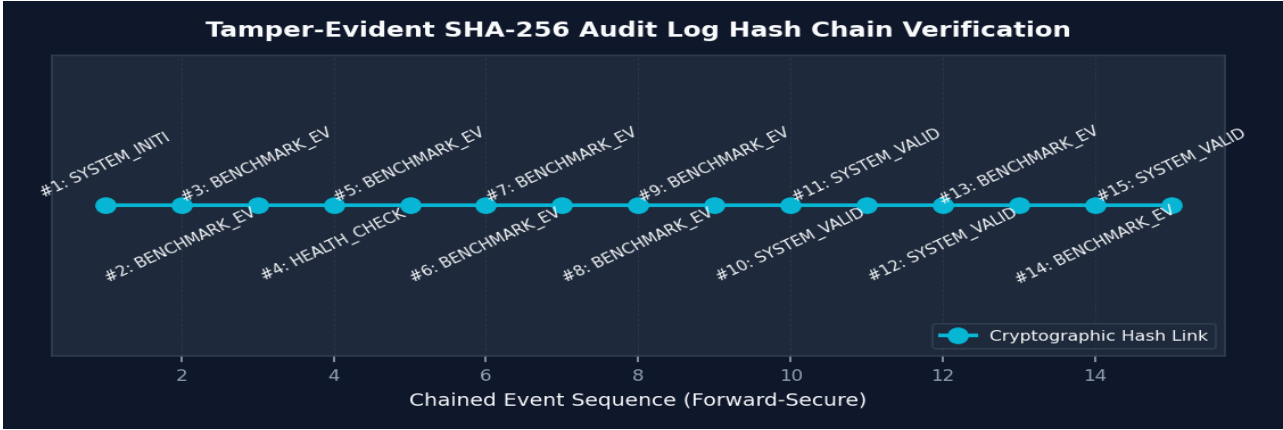
## Finding Severity Breakdown



Domain Shift & Statistical Drift Comparison

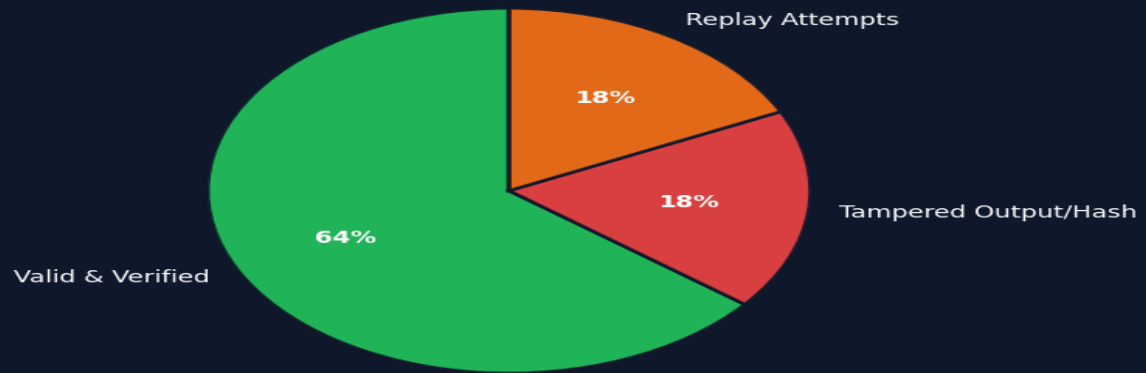


Audit Log Hash Chain Timeline



Inference Provenance Cryptographic Status

## Cryptographic Inference Provenance Status



Detailed Security & Integrity Findings (Evidence-Driven)

| ID                    | Category     | Sev.     | Conf. | Risk  | Disposition | Evidence & Telemetry                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|--------------|----------|-------|-------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F-DS-BENCHMARK-001    | NEAR-DUPLICA | HIGH     | 95%   | 17.9  | ACCEPT      | <b>Perceptual Near-Duplicate Flooding Detected (1 clusters)</b> <ul style="list-style-type: none"><li>Detected 1 duplicate cluster(s) with similarity &gt;= 0.95</li><li>Total duplicate samples: 5</li></ul>                                                                                                                                                                                               |
| F-DS-BENCHMARK-002    | RELABEL      | HIGH     | 92%   | 29.2  | ACCEPT      | <b>Systematic Mislabelling &amp; Label Noise Detected (5 samples)</b> <ul style="list-style-type: none"><li>5 samples conflict with nearest visual feature neighbors.</li><li>Affected sources: ['source_rogue_3']</li></ul>                                                                                                                                                                                |
| F-DS-BENCHMARK-003    | OOD          | HIGH     | 75%   | 19.4  | ACCEPT      | <b>Out-of-Distribution Inliers/Outliers Detected (4 samples)</b> <ul style="list-style-type: none"><li>4 samples exceeded anomaly threshold.</li><li>Overall OOD score: 0.37</li></ul>                                                                                                                                                                                                                      |
| F-DS-BENCHMARK-004    | TRIGGER      | CRITICAL | 95%   | 74.4  | QUARANTINE  | <b>Potential Visual Trigger / Backdoor Patch Identified (1 clusters)</b> <ul style="list-style-type: none"><li>Identified 1 localized trigger pattern cluster(s).</li><li>Affected samples: ['sample_0049', 'sample_0048', 'sample_0046', 'sample_0047', 'sample_0045']</li></ul>                                                                                                                           |
| F-DS-BENCHMARK-005    | SOURCE POISO | HIGH     | 90%   | 49.7  | QUARANTINE  | <b>Source/Contributor Poisoning Anomaly (2 sources flagged)</b> <ul style="list-style-type: none"><li>Flagged poisoned sources: ['source_rogue_3', 'source_flood_2']</li></ul>                                                                                                                                                                                                                              |
| F-MOD-SUB-ONN-001     | MODEL_SUBSTI | CRITICAL | 100%  | 100.0 | QUARANTINE  | <b>Model Integrity Deviation: SUBSTITUTION DETECTED</b> <ul style="list-style-type: none"><li>Candidate SHA-256: 82bd8ad3fde3b94e2024ff54f3743a1e98b0d07004752c6d63a0c51493ba313f</li><li>Reference SHA-256: 8bb12146a821fa67389d3160a218330e51c43e55bd734954f3422414834f78c1</li><li>Parameter difference ratio: 82%</li></ul>                                                                             |
| F-MOD-SUB-ONN-002     | BACKDOOR_BEH | HIGH     | 90%   | 63.2  | REVIEW      | <b>Behavioural Battery Instability Detected (Agreement: 0%)</b> <ul style="list-style-type: none"><li>Behavioural class agreement: 0%</li><li>Anomalous test inputs: 24</li></ul>                                                                                                                                                                                                                           |
| F-TAMP-58DFF5         | TAMPERING    | CRITICAL | 100%  | 100.0 | QUARANTINE  | <b>Inference Output Tampering Detected in Record REC-D65C06E75EDC-TAMP</b> <ul style="list-style-type: none"><li>OUTPUT_PAYLOAD (expected a166afe5cca2..., got a6e8c128ab8c...)</li></ul>                                                                                                                                                                                                                   |
| F-REP-94D233          | REPLAY       | CRITICAL | 100%  | 100.0 | QUARANTINE  | <b>Inference Replay Attack Detected: Nonce Reuse in Record REC-RPLY-FDC7474D-REPLAY</b> <ul style="list-style-type: none"><li>Cryptographic nonce 'nonce_auth_001' was previously used in record 'REC-BED19CBA52D8'.</li><li>Monotonic sequence number 101 already consumed by record 'REC-BED19CBA52D8'.</li><li>Sequence regression: submitted sequence 101 is &lt;= current sequence head 101.</li></ul> |
| F-DS-SHIFT-DOMAIN-001 | OOD          | MEDIUM   | 75%   | 8.1   | ACCEPT      | <b>Out-of-Distribution Inliers/Outliers Detected (2 samples)</b> <ul style="list-style-type: none"><li>2 candidate samples exceeded the 95th anomaly percentile.</li><li>Overall population OOD anomaly score is 0.45.</li><li>Evaluation method: IsolationForest (Internal Population).</li></ul>                                                                                                          |
| F-RPLY-FDE796         | REPLAY       | HIGH     | 95%   | 61.1  | QUARANTINE  | <b>Adversarial Replay Attempt Detected on Record REC-RPLY-FDC7474D-REPLAY</b> <ul style="list-style-type: none"><li>Duplicate record_id 'REC-RPLY-FDC7474D-REPLAY' already registered in database.</li><li>Cryptographic nonce 'nonce_auth_001' was previously used in record 'REC-BED19CBA52D8'.</li><li>Monotonic sequence number 101 already consumed by record 'REC-BED19CBA52D8'.</li></ul>            |

Assurance Threat Coverage Statement

| ID | Attack / Threat Capability | Status    | Access Mode      | Primary Method                            | Key Limitation                            |
|----|----------------------------|-----------|------------------|-------------------------------------------|-------------------------------------------|
| T1 | Label Flipping Detection   | SUPPORTED | Dataset Labels + | Feature k-NN neighborhood consensus & ... | Requires distinct feature cluster sepa... |
| T2 | Systematic Mislabelling    | SUPPORTED | Dataset Metadata | Hierarchical contributor & batch anoma... | Effectiveness depends on presence of s... |

|     |                                       |                     |                  |                                           |                                           |
|-----|---------------------------------------|---------------------|------------------|-------------------------------------------|-------------------------------------------|
| T3  | Duplicate Flooding                    | SUPPORTED           | Image Files (Bla | Perceptual Hashing (pHash, dHash) Hamm... | Sensitivity tuned via threshold; heavi... |
| T4  | Out-of-Distribution (OOD) Injection   | SUPPORTED           | Dataset or Refer | Isolation Forest & statistical feature... | High-dimensional semantic outliers req... |
| T5  | Trigger / Backdoor Injection          | PARTIALLY_SUPPORTED | Dataset Images   | Localized patch spatial consistency & ... | Blended or frequency-domain watermarks... |
| T6  | Model Substitution                    | SUPPORTED           | Model File (.onn | Cryptographic SHA-256 identity & param... | Requires trusted reference fingerprint... |
| T7  | Model Weight Modification             | SUPPORTED           | Model Weights (W | Tensor statistics, layer shapes, and m... | External weights or opaque black-box r... |
| T8  | Backdoor Behaviour Testing            | SUPPORTED           | Inference Execut | Controlled perturbation battery (brigh... | Cannot prove absence of dormant trigge... |
| T9  | Inference Record Tampering            | SUPPORTED           | Full Record Payl | Cryptographic hash binding + Ed25519 d... | Protects recorded inferences; requires... |
| T10 | Inference Replay Attacks              | SUPPORTED           | Inference Log St | Cryptographic nonces, monotonic sequen... | Requires database persistence of past ... |
| T11 | Provenance Configuration Substitution | SUPPORTED           | Provenance Metad | Canonical JSON serialization hashing f... | Config schema changes must remain back... |
| T12 | Distribution Shift & Drift            | SUPPORTED           | Reference & Cand | PSI, 2-sample KS test, Wasserstein dis... | Requires sufficient sample volume (N >... |

## System Limitations & Air-Gap Compliance

- Framework runs 100% locally and does not transmit data or consult cloud foundation models.
- Statistical heuristics cannot mathematically prove total absence of steganographic or adversarial latent triggers.
- Model weight analysis on ONNX models depends on initializers embedded within the graph file.
- Offline cryptographic protection guarantees tamper-evident provenance but assumes private key is stored securely in air-gapped host.